Exploring the API of 4TU.ResearchData

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4TU.ResearchData
SCIENCE • ENGINEERING • DESIGN

Introduction

01 What are APIs

02 Common language of Web APIs

03 Why APIs are relevant for research

04 Using the Unix terminal to work with Web APIs

05 The Web API of 4TU.ResearchData

01 What are APIs

Introduction to APIs

1. API (Application Programming Interface): A bridge that allows different software systems to communicate.
2. APIs define how software components interact.

Client
End User with Browser
API
Server
Server Back-end System
Request
Response
API

Make the Order
Delivery of order
Take the Order
Bringing from Kitchen
Customer
Waiter
Chef

Introduction to APIs

02 Common language for APIs

1. Web API is an API that allows different software systems to communicate via the web, using HTTP protocol.
2. They use a **shared vocabulary** of methods: GET, POST, PUT, DELETE to define how to exchange **resources** across the web.
3. Resources are something the API makes available and lets you interact with, e.g. (datasets, authors, collections).
4. Most Web APIs today use the JSON format to exchange data.

Introduction to APIs

03 Why APIs are relevant for research

1. (In general) APIs enable automation, integration, and reuse of data and services.
2. Enable programmatic access to research data (no manual downloads)
3. Facilitate reproducibility and automation of workflows
4. Support interoperability across systems and tools

Example: Retrieve datasets via API → analyze in Python/R → publish results

Introduction to APIs

04 Using the Unix terminal with REST APIs

1. The Unix terminal allows HTTP requests using `***curl***`.

`***curl***` = client URL: send requests and receive responses.

Examples:
curl [options] [URL]

→ Request → Server → JSON response

Introduction to APIs

05 The Web API of 4TU.ResearchData

[base URL] : https://data.4tu.nl

[specific resource URL:endpoint]:
https://data.4tu.nl/v2/articles

List of *(documented)* endpoints:
https://djehuty.4tu.nl/

Exercise #1 : Checking the connection with the server using the web browser and CURL in the Unix terminal

- Open a web browser and write: https://data.4tu.nl/v2/articles
- One the Unix terminal and write:
  `curl –X GET https://data.4tu.nl/v2/articles | jq
- How are both responses compared to each other?